

FVM Foundations · free cheat sheet · page 1: the method

CFD by Hand

The finite volume method from scratch: cells, faces, integrating over a control volume, and the cell equation.

Mesh and integrals

Topic	Equation	Remember
1D mesh	$\Delta x = L/n, \quad x_{P,i} = (i - \frac{1}{2})\Delta x, \quad x_{w,i} = (i-1)\Delta x$	n cells, $n+1$ faces
2D mesh	cells $N_x N_y$; faces $(N_x+1)N_y + N_x(N_y+1)$	interior face shared by 2 cells
Names	P this cell; W E S N neighbour centres; w e s n faces	capitals centres, small faces
Cell volume	1D: $dV = A\,dx$, $\Delta V = \Delta x$ ($A=1$); 2D: $\Delta V = \Delta x \Delta y$	per unit area / depth
Midpoint rule	$\int_{\Delta V} f\,dV \approx f_P\,\Delta V$	exact for straight lines; error $f''\Delta x^3/24$
Derivatives	$\int_w^e \frac{dJ}{dx}dx = J_e - J_w, \qquad \int_{\Delta V} \nabla \cdot \mathbf{J}\,dV = \sum_f \mathbf{J}\cdot\hat{\mathbf{n}}\,A_f$	interior fluxes cancel: conservative

Face approximations

Quantity	Formula	Remember
Face slope	$\left(\frac{d\phi}{dx}\right)_e \approx \frac{\phi_E - \phi_P}{\delta x_{PE}}$	second order: error $\propto \delta x^2$
Face value	$\phi_e \approx \frac{1}{2}(\phi_P + \phi_E)$; uneven: $(1-f)\phi_P + f\phi_E$, $f = \frac{x_e - x_P}{x_E - x_P}$	central differencing
Boundary face	$\left(\frac{d\phi}{dx}\right)_A \approx \frac{\phi_P - \phi_A}{\Delta x/2}$	half a cell, not a whole one
Upwind value	$\phi_e = \phi_P$ if $F_e > 0$, else ϕ_E	take the upstream cell

The four steps for every cell

Step	1D diffusion example	Remember
1 Equation	$\frac{d}{dx}\left(\Gamma\frac{d\phi}{dx}\right) + S = 0$	
2 Integrate	$\left(\Gamma\frac{d\phi}{dx}\right)_e - \left(\Gamma\frac{d\phi}{dx}\right)_w + S\Delta V = 0$	exact so far
3 Approximate	$D_e(\phi_E - \phi_P) - D_w(\phi_P - \phi_W) + S\Delta V = 0, \quad D = \Gamma/\delta x$	the only approximation
4 Collect	$a_P\phi_P = a_W\phi_W + a_E\phi_E + S_u, \quad a_P = a_W + a_E - S_p$	one per cell
Excel	$\phi_P = (a_W\phi_W + a_E\phi_E + S_u)/a_P$, iterative calc on	or `MINVERSE` matrix solve

Boundaries, sources, meshes and convection

Boundary cells (zero the missing a_{nb}, add to S_u, S_p)

Boundary	S_u and S_p	Remember
Fixed value	$S_u = 2D\phi_b, \quad S_p = -2D$	the 2 is the half cell
Known flux q in	$S_u = qA, \quad S_p = 0$	flux out: $-qA$
Zero gradient	nothing	insulated, symmetry, outlet
Mixed (Robin)	$S_u = h_c\phi_\infty,\ S_p = -h_c, \quad h_c = \dfrac{1}{1/h + (\Delta x/2)/\Gamma}$	$h \to \infty$: fixed value
Source	$S = S_C + S_P\phi$: $S_u \mathrel{+}= S_C\Delta V,\ S_p \mathrel{+}= S_P\Delta V$	keep $S_P \le 0$

Meshes

Case	Coefficient	Remember
Uneven cells	$a_W = \Gamma/\delta x_{WP}, \quad a_E = \Gamma/\delta x_{PE}$	each face its own D
Two materials	$D_f = \dfrac{1}{\delta x_{Pf}/\Gamma_P + \delta x_{fN}/\Gamma_N}$	add resistances, never average Γ
2D rectangles	$a_E = a_W = \Gamma\Delta y/\Delta x, \quad a_N = a_S = \Gamma\Delta x/\Delta y$	face length / centre distance
Any cell shape	$\sum_f \Gamma A_f \dfrac{\phi_N - \phi_P}{d_{PN}} + SV_P = 0, \quad a_N = \dfrac{\Gamma A_f}{d_{PN}}$	one a_{nb} per face
Triangle	area $\frac{1}{2}\lvert x_1(y_2-y_3)+x_2(y_3-y_1)+x_3(y_1-y_2)\rvert$, centroid = mean of corners	orthogonal mesh assumed

Convection-diffusion ($F = \rho uA$, $D = \Gamma A/\delta x$, $Pe = F/D$)

Scheme	Coefficients	Boundaries
Equation	$F_e\phi_e - F_w\phi_w = D_e(\phi_E - \phi_P) - D_w(\phi_P - \phi_W)$	$a_P = \sum a_{nb} + \Delta F - S_p$
Central	$a_W = D + F/2, \quad a_E = D - F/2$	inlet $(2D+F)\phi_A$; outlet $(2D-F)\phi_B$
Upwind	$a_W = D + \max(F, 0), \quad a_E = D + \max(-F, 0)$	inlet $(2D + F)\phi_A$; outlet $2D\phi_B$
2D upwind	$a_S = D_s + \max(F_s, 0), \quad a_N = D_n + \max(-F_n, 0)$	wall $2D\phi_w$; outlet nothing
Exact 1D	$\phi = \phi_A + (\phi_B - \phi_A)\dfrac{e^{Pe\,x/L} - 1}{e^{Pe} - 1},\ Pe = \rho uL/\Gamma$	central wiggles if cell $Pe > 2$